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Task Description:

While ultrasound has proved invaluable for a wide variety of medical assessments during spaceflight, it is largely blind to two important aspects of human physiology: blood oxygenation and blood volume. Near-infrared spectroscopy (NIRS), in contrast, is highly sensitive to blood oxygenation and volume, and hence has multiple medical uses ranging from the gold standard of pulse oximetry, to tissue oxygenation and perfusion, to imaging of brain function, hemorrhage, or ischemia. Currently, the only related measurement available in flight is pulse oximetry, and hence NIRS could be a key complement to ultrasound for spaceflight medical use. Importantly, a single NIRS system could support all of the above applications, while using non-ionizing near-infrared light, and remaining non-invasive, low power, and highly portable. Existing NIRS systems, however, are ill-suited to spaceflight (large, heavy, and consume too much power) and many are also incapable of imaging.
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Our objective of this project was to develop and test a new NIRS system, NINscan-M: a multi-use, mobile device with the following capabilities: (1) imaging of regional tissue hemodynamics, including brain function, (2) point measurements of oxygenation suitable for muscle assessment, and (3) tissue oximetry. NINscan-M is an embedded microcontroller system, based on our laboratory's series of prototype non-imaging NIRS devices designed for mobile and long-duration monitoring. In addition to the NIRS capabilities above, the system also includes the ability to record up to 12 synchronized, auxiliary data streams (both analog and digital) for monitoring cardiac, brain, and skeletal muscle electrical activity, motion, force production, and temperature to further enhance its domain of application. These capabilities can fill various key roles in spaceflight medical assessment, and could help address gaps in at least eight Human Research Roadmap (HRP) – Integrated Research Plan (IRP) risk categories: spaceflight-induced intracranial hypertension, recognizing injured or ill crewmembers, muscle status monitoring, orthostatic intolerance assessment, detection or monitoring of behavioral or psychiatric conditions, effects of radiation on the central nervous system, sleep loss, and cardiac effects of spaceflight.

NINscan-M was designed to enable NIRS as well as auxiliary sensors to be flown and used singly or in combination, as indicated by medical operations requirements. It is operated by a single on/off switch, and a Bluetooth wireless communication capability was integrated to enable real-time monitoring of data streams. Human testing of the system demonstrated: (1) the ability to simultaneously acquire 64-channel NIRS imaging data along with electrocardiography (ECG), electromyography (EMG) and electro-oculography (EOG), tri-axial accelerometry, force, and temperature data; (2) imaging of muscle oxygen utilization and blood volume with synchronized measurement of force production during isometric muscle contractions to help assess muscle strength and endurance; and (3) imaging regional changes in brain activity in lateral prefrontal cortex associated with working memory tasks. NINscan-M thus provides new capabilities, complementary to those currently available in-flight, which are important for medical assessments across a wide variety of spaceflight medical conditions.

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Our central objective is to develop and test a new NIRS system, NINscan M: a multi-use, mobile device with three distinct NIRS capabilities: (1) imaging of regional tissue oxygenation, perfusion and hemodynamics, including brain function, (2) point measurements of oxygenation suitable for muscle assessment, and (3) tissue oximetry.

The NINscan M is an embedded microcontroller system, based on our laboratory's series of prototype non-imaging NIRS devices designed for mobile and long-duration monitoring. In addition to the three NIRS capabilities above, the system also includes the ability to record up to 12 synchronized, auxiliary data streams for monitoring cardiac, brain, and skeletal muscle electrical activity, motion, force production, and temperature to further enhance its diagnostic value. These capabilities can fill various key roles in spaceflight medical assessment, and could help address gaps in at least eight Human Research Roadmap risk categories: spaceflight-induced intracranial hypertension, recognizing injured or ill crewmembers, muscle status monitoring, orthostatic intolerance assessment, detection or monitoring of behavioral or psychiatric conditions, effects of radiation on the central nervous system, sleep loss, and cardiac effects of spaceflight. The system design will enable the NIRS and auxiliary sensors to be flown and used singly or in combination, as indicated by medical operations requirements. To minimize the data management burdens for astronauts, NINscan M will be designed to seamlessly integrate with the National Space Biomedical Research Institute (NSBRI)-supported SmartMED platform for automatically detecting, consolidating, storing, communicating, and displaying biomedical and environmental data. Once developed, we will thoroughly characterize the NINscan M system, and perform human testing for: (1) sensitivity and specificity to changes in blood volume and oxygenation when imaging skin perfusion (shallow layers) and imaging brain function (deep tissue), and (2) the ability to quantify muscle force production and oxygen extraction to assess muscle performance.

During development, we will continue working with International Space Station Medical Project (ISSMP) Element to ensure the resulting system is designed and developed within the constraints and requirements for spaceflight hardware and software. NINscan M will thus provide new capabilities, complementary to those currently available in-flight, which are important for medical assessments across a wide variety of spaceflight medical conditions.

Rationale for HRP Directed Research:

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Research Impact: Currently, pulse oximetry is available onboard the International Space Station (ISS), and one would expect it to be flown on future exploration class missions. A pulse oximeter, however, provides only a small subset of the information available from a NIRS measurement. In particular, pulse oximetry can only provide oxygenation measures from arterial blood in superficial tissue when a sufficiently strong pulse is available, whereas NIRS measurements can provide surface and deep arterial measurements as well as venous and whole-tissue measurements. Moreover, NIRS does not require clear pulse signals, being usable in compartment syndrome, during muscle contractions, or in patients with a weak or thready pulse. A NIRS-based imager can further provide spatial information about tissue oxygenation or perfusion. This could support: (1) assessing the relative conditioning status of (and optimizing training for) different muscle groups, (2) assessing cerebral hemodynamics for conditions such as visual impairment/intracranial pressure (VIIP), including tissue oxygenation in different tissue layers (e.g., scalp versus cerebral oxygenation, or skin versus fat versus deeper muscle layers), (3) sleep physiology studies, or even (4) identifying the location or evaluating the size of an internal hemorrhage. When deploying a sufficient number of near-infrared wavelengths and source/detector locations, one can further measure water content (e.g., edema), and correct measurements of deeper tissue layers for skin color, fat layers, and dynamic changes in systemic physiology

Research Impact/Earth Benefits:

(cardiac, respiratory, and other vasomotor interference). These capabilities of NIRS over standard pulse oximetry are not currently available in spaceflight but could be provided by NINscan-M. The oxygenation and perfusion measures just described can also be enhanced by various auxiliary measurements. For example, electrocardiography (ECG) can be combined with systemic NIRS measurements to help estimate cardiac output. Electromyography (EMG) can be combined with muscle oxygenation measures to better understand muscle endurance and conditioning. Accelerometry and temperature measures can help identify and, when needed, compensate for environmental influences on NIRS or other physiological/auxiliary measurements. The goal with NINscan-M was to provide not only shallow- and deep-tissue NIRS imaging capabilities, but also the ability to support recording of various physiological measures—each useful in their own right as well as to enhance NIRS assessment capabilities. NIRS and auxiliary functions are provided modularly, so that missions with different requirements need only pack and deploy the necessary components.

Earth Benefits: The same capabilities that are useful for exploration spaceflight are also relevant for Earth-based applications. For example, continuous non-invasive, long-duration brain monitoring for cerebral hemorrhage is key and an unmet need following neurosurgery, stroke, and traumatic brain injury. Non-invasive monitoring of brain motion within the skull is a novel capability useful both in research and prevention of traumatic brain injury, and our group is working on a project supported by the National Football League Players Association on this topic. Real-time muscle oxygenation imaging can be used to optimize elite athlete training as well as enhance rehabilitation protocols. And the entire field of human neuroscience is largely constrained to using large machines for brain function monitoring that require the subject to remain as still as possible. NINscan-M provides the ability to monitor brain function during people's daily activities, opening up entirely new domains of investigation. The NINscan-M device is expected to be useful in these and other contexts.

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Research Impact: Currently, pulse oximetry is available onboard the ISS, and one would expect it to be flown on future exploration class missions. A pulse oximeter, however, provides only a small subset of the information available from a full NIRS measurement. In particular, pulse oximetry can only provide oxygenation measures in arterial blood when a sufficiently strong pulse is available, whereas NIRS measurements can provide arterial measurements as well as venous and whole-tissue measurements, and does not require clear pulse signals (e.g., compartment syndrome, during muscle contractions, or in patients with a weak or thready pulse). A NIRS-based imager can further provide spatial information about tissue oxygenation or perfusion. This can help, for example, with: (1) assessing cerebral hemodynamics for conditions such as visual impairment/intracranial pressure (VIIP), including tissue oxygenation in different tissue layers (e.g., scalp versus cerebral oxygenation, or skin versus fat versus deeper muscle layers), (2) identifying the location or evaluating the size of an internal hemorrhage, or (3) assessing the relative conditioning status of different but adjacent muscle groups. Moreover, when deploying a sufficient number of near-infrared wavelengths and source/detector locations, one can further measure water content (e.g., edema), and correct measurements of deeper tissue layers for skin color, fat layers, and dynamic changes in systemic physiology (cardiac, respiratory, and other vasomotor interference). These capabilities of NIRS over standard pulse oximetry are not currently available in spaceflight but could be provided by NINscan M. The oxygenation and perfusion measures just described can also be enhanced by various auxiliary measurements. For example, electrocardiography (ECG) can be combined with systemic NIRS measurements to help estimate cardiac output. Electromyography (EMG) can be combined with muscle oxygenation measures to better understand muscle endurance and conditioning. Accelerometry and temperature measures can help identify and compensate for environmental influences on NIRS measurements. Our goal with NINscan M is to provide not only shallow- and deep-tissue NIRS imaging capabilities, but also the ability to record various auxiliary measures—each useful in their own right—to enhance NIRS assessment capabilities. NIRS and auxiliary functions are being provided modularly, so that missions with different requirements need only pack and deploy the necessary components. Importantly, the system will integrate with the SmartMED platform to minimize astronaut time and training burdens arising from data collection and management.

Earth Benefits: The same capabilities that are useful for exploration spaceflight are also relevant for Earth-based applications. For example, continuous non-invasive brain monitoring for cerebral hemorrhage is a key and unmet need following neurosurgery, stroke, and traumatic brain injury. Non-invasive monitoring of brain motion within the skull is another unmet need both in research and prevention of traumatic brain injury. Real-time muscle oxygenation imaging can be used to optimize elite athlete training as well as enhance rehabilitation protocols. And the entire field of human neuroscience is largely constrained to using large machines for brain function monitoring that require the subject to remain as still as possible. NINscan-M provides the ability to monitor brain function during people's daily activities, opening up entirely new domains of inquiry. The NINscan-M device is expected to be useful in these and other contexts.

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We achieved our overall objective by completing and testing our NINscan-M v2 prototype. Major achievements were:

NINscan-M Development

• NINscan-M Prototype: NINscan-M is the smallest, lightest, lowest-power and easiest to use NIRS hemodynamic imaging system available. Our modular design enables flexible deployment for spaceflight or analog missions and a broad range of Earth-based clinical and research applications.

• NIRS Sensors: We developed and tested 3 sensors: (i) a large, 64-channel imaging sensor optimized for broad-area strip imaging, (ii) a more compact 64-channel imaging sensor optimized for regional imaging, and (iii) a quad point sensor that can monitor 4 separate sites simultaneously.

• Biopotential Inputs: The 8-channel analog input provides flexible biopotential recording with 24-bit resolution at 250 Hz, including simultaneous acquisition of multi-channel ECG, EMG, and/or EOG, as desired.

• Auxiliary Inputs: Four user-input buttons, a tri-axial accelerometer, a temperature sensor, and a force sensor (sensitive up to 100-lb) were also incorporated.

• Power: Power can be switched between AA batteries and A/C adapter plug-in.

• GUI: A graphical user interface was developed for monitoring the NINscan-M Bluetooth data for real-time interactive recordings and data quality assurance applications. NINscan-M was also integrated with our separate SpaceMED data acquisition system, allowing real-time NINscan-M data management.

Task Progress:	• Ease-of-Use Features: (i) single-switch (on/off) system operation, (ii) automatic gain-control built into the power-on sequence to optimize signals for all NIRS channels, (iii) modular (pluggable) auxiliary channels enable selection of only those sensors required for any given application, and (iv) Bluetooth communication along with developed GUI enable real-time data monitoring and quality assurance. NINscan-M Demonstrations • Oxygenation changes during muscle contraction were imaged while measuring EMG and force production. Such data can be used to assess muscle endurance in multiple muscles simultaneously, as well as muscle strength assessments. • Imaging of low-amplitude, regional oxygenation changes during functional brain activation was achieved. • Demonstrated low-noise 9-lead ECG measurements at 250Hz. • Demonstrated simultaneous multi-channel ECG, EMG, and EOG recordings. • Demonstrated synchronized accelerometry, force, temperature, and user-button inputs. Additional Achievements • Tested the NINscan technology in a parabolic flight campaign (4 flights; 19 subjects). • Published the first 24-hr continuous brain hemodynamic monitoring, using the NINscan technology. • Conducted and published the most detailed study to date investigating the sensitivity of NIRS to brain tissue. • Developed novel data analysis techniques for NIRS data to help correct deep-tissue measures for skin color and fat layers. • Discovered the sensitivity of our NIRS instruments to intracranial brain motion. ANNUAL REPORTING IN OCTOBER 2014 We completed most of our first functioning NINscan-M recorder (v1) in year 1, providing the first truly wearable 64-channel NIRS imaging system. During Year 2, we completed it and focused on testing v1 and a developing a second-generation device (v2) to miniaturize the system and improve sensitivity and functionality. The v2 system is now fully designed and boards are being fabricated and populated. Important achievements during the 2nd year of the project include: Enhanced NIRS sensitivity: Automatic gain-control has been incorporated for both the light sources and detectors. This will increase the dynamic range of the system 20x to 60x, which is a key feature for collecting suitable data for imaging applications. We have also redesigned the power supply circuit to reduce noise and to enable power-supply switching between untethered battery use and wall-power. Imaging Sensors: Two new sensors are being fabricated, including an optimized 64-channel imaging sensor, and a 4-arm, multi-site sensor. The imaging sensor will optimally utilize the system's dynamic range for higher sensitivity and higher resolution imaging. The multi-site sensor will enable monitoring of multiple regions of the body simultaneously (e.g., bilateral brain assessment, or simultaneous head, arm and leg measurements). Auxiliary Inputs: The 8-channel analog input enables recording of a wide range of signals (e.g., EMG, ECG, and/or EEG). These are collected with 16-bit resolution at 250Hz, just like NIRS data. A tri-axial digital accelerometer, a digital temperature module, and a force sensor used to sense force up to 100lb have also been incorporated into the system. User Interface: The Bluetooth module was activated and tested for wireless data streaming. This was then integrated with our separate SpaceMED data acquisition system, allowing real-time viewing and archiving of NINscan M data as it is streamed off via a wireless Bluetooth connection. Analog Testing: To identify any operational issues or constraints for such multi-modal monitoring, we tested the base NINscan technology during a parabolic flight campaign, including simultaneous NIRS, ECG, and accelerometry. This allowed us to assess the feasibility of using NINscan in analog/operational settings, to evaluate the performance of our technology in-flight versus on the ground, and to assess the sensitivity of our technology to cerebral alterations associated with gravitational changes. NINscan v1 represents the smallest, lightest, and lowest-power NIRS imaging system currently in existence, and v2 will be approximately a third the size of v1. While further characterization, human testing, and validation will occur in year 3, the flexible design will enable deployment in a variety of settings or for a variety monitoring needs. These range from general medical risks, to intracranial pressure risks, musculoskeletal risks, radiation risks, behavioral and performance risks, and cardiovascular risks.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
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